

AZURE AGCM, AB, Canada

WMO: 719810

Lat: 50.5119N

Lon: 114.0131W

Elev: 3750

StdP: 12.81

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.8	-13.5	-24.8	1.6	-18.6	-19.3	2.2	-13.1	24.5	36.8	21.2	35.8	5.8	240	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	83.8	59.4	80.4	58.9	77.1	57.6	64.1	75.6	62.1	74.4	60.3	73.0	7.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.9	88.6	69.9	57.4	80.8	67.2	55.3	74.7	65.0	31.3	76.0	29.7	74.3	28.4	72.8	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.6	15.9	-25.6	90.2	5.0	4.1	-29.2	93.2	-32.1	95.6	-34.9	97.9	-38.5	100.9		
(5)			-25.9	68.5	4.9	2.8	-29.5	70.5	-32.3	72.1	-35.1	73.6	-38.7	75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	22.3	19.0	28.7	38.6	48.8	55.9	61.2	60.5	52.8	40.6	28.9	18.9
	DBStd	18.93	15.13	15.40	14.32	8.89	8.00	5.64	5.01	5.80	9.17	9.77	13.57	15.11
	HDD50	4868	861	867	662	351	120	12	1	3	75	314	637	965
	HDD65	9247	1325	1287	1126	791	502	279	137	159	372	756	1083	1430
	CDD50	1148	1	0	1	10	84	188	348	329	160	23	4	0
	CDD65	51	0	0	0	0	1	4	19	20	7	0	0	0
	CDH74	1445	0	0	0	8	73	118	496	533	214	3	0	0
	CDH80	329	0	0	0	0	6	20	116	144	43	0	0	0
	WSAvg	7.0	8.0	6.8	7.6	7.8	7.7	7.1	5.5	5.2	6.3	7.0	7.9	7.7
	PrecAvg	23.5	1.1	1.0	1.4	1.9	3.1	4.2	2.4	2.4	2.0	1.3	1.3	1.1
(15) Precipitation	PrecMax	36.7	2.0	1.5	3.5	3.6	5.7	14.9	6.3	5.2	5.7	2.4	3.2	2.5
	PrecMin	16.0	0.4	0.2	0.6	1.1	0.8	0.7	0.6	0.4	0.2	0.2	0.1	0.4
	PrecStd	4.3	0.4	0.4	0.7	0.7	1.4	2.7	1.5	1.1	1.3	0.6	0.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	54.8	59.7	73.5	80.5	83.1	87.8	88.2	84.9	71.0	64.7	52.2
		MCWB	40.8	41.3	43.4	49.6	53.5	61.4	61.7	58.7	56.2	50.0	45.7	38.3
	2%	DB	48.6	48.5	55.9	65.7	75.8	77.1	83.4	83.9	80.2	65.7	56.0	46.2
		MCWB	38.5	37.5	41.4	46.4	53.0	57.1	61.4	58.9	55.7	48.8	43.0	36.9
	5%	DB	45.1	44.0	51.7	60.0	70.9	73.4	79.5	80.0	75.6	60.8	51.2	42.6
		MCWB	36.2	34.6	39.2	43.2	51.6	56.2	61.4	59.1	54.7	46.6	40.5	34.8
	10%	DB	42.0	39.8	47.3	54.7	66.0	70.0	75.9	76.0	70.2	56.4	46.7	39.5
		MCWB	34.5	32.2	37.3	40.7	48.9	54.6	60.3	58.1	52.7	43.8	37.8	32.2
	0.4%	WB	41.7	42.2	45.4	50.2	56.3	64.3	68.6	65.1	59.8	52.7	47.2	40.4
		MCDB	52.7	52.4	57.7	71.5	74.6	75.2	79.3	76.5	76.0	67.0	61.0	49.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.5	38.0	42.2	46.9	53.8	60.7	65.1	62.6	57.6	49.8	44.1	37.1
		MCDB	47.2	47.5	53.5	63.4	72.6	73.3	75.8	75.9	74.2	62.6	55.7	44.7
	5%	WB	36.5	35.1	40.0	44.1	52.0	57.9	63.2	60.9	55.6	47.2	40.9	34.8
		MCDB	44.4	43.0	51.0	58.2	68.1	70.0	74.7	74.8	71.3	59.6	50.2	42.3
	10%	WB	34.7	32.6	37.4	41.4	50.0	55.8	61.5	59.2	53.6	44.5	37.8	32.5
		MCDB	41.9	39.2	47.3	52.7	63.9	67.1	73.3	72.9	69.2	55.7	46.0	39.1
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	19.7	20.1	22.0	23.3	22.6	25.3	26.1	24.5	21.6	19.1	18.6
		MCDBR	20.3	23.7	25.4	32.3	32.4	29.1	32.7	33.9	33.2	27.7	23.9	20.2
		MCWBR	13.4	14.9	14.5	16.1	14.6	13.9	15.7	14.3	14.1	14.1	14.1	13.7
	5% WB	MCDBR	19.9	23.1	24.6	30.6	28.6	25.6	27.6	28.4	29.8	26.9	23.6	20.5
		MCWBR	13.5	15.4	14.5	15.8	13.5	13.5	15.3	14.3	13.6	14.2	14.4	14.3
(40) Clear-Sky Solar Irradiance	taub		0.251	0.269	0.281	0.335	0.341	0.340	0.338	0.354	0.312	0.277	0.254	0.234
	taud		2.441	2.436	2.464	2.321	2.376	2.416	2.430	2.386	2.483	2.555	2.508	2.446
	Ebn at Noon		259	281	297	288	290	291	289	279	281	276	257	250
	Edh at Noon		18	24	28	36	36	35	34	34	28	21	17	16
(44) All-Sky Solar Radiation	RadAvg		370	676	1058	1487	1744	1879	2100	1712	1204	744	397	285
	RadStd		28	54	78	99	136	138	134	118	122	78	33	21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page